

Notes: Rajan, Seru, and Vig (JFE, 2015)

The Failure of Models that Predict Failure: Distance, Incentives, and Defaults

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Why this market is the right laboratory	4
2.2	Primary data set: securitized subprime mortgages	4
2.3	What the primary data observe	5
2.4	Secondary data: securitization rates and the NCFC loan file	5
2.5	Default and other core outcome variables	6
2.6	Descriptive facts	6
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Conceptual framework: low- versus high-securitization regimes	7
3.2	Three empirical predictions of the framework	7
3.3	Main interest-rate regression on hard information	8
3.4	Shrinkage test for the distribution of interest rates	8
3.5	NCFC pricing and acceptance tests using the internal rating	9
3.6	Contemporaneous default regressions on the interest rate	9
3.7	Baseline default model estimated in the low-securitization regime	10
3.8	Out-of-sample prediction errors in the high-securitization period	10
3.9	Rolling-window, placebo, and house-price-augmented designs	10
4	Identification and empirical design (what makes the estimates credible)	11
4.1	What the paper identifies	11
4.2	Why the pricing evidence is not just lender composition	11
4.3	Why the story is about lost information, not FICO becoming more informative	12
4.4	Why the forecasting failure is not just generic misspecification	12
4.5	Cross-sectional designs that isolate the information channel	12
4.6	Alternative explanations the paper addresses	13
4.7	Role of investors and limits of interpretation	13
4.8	Relation to the literature	14
5	Tables and figures	14
5.1	Tables 1 and 2: market growth and securitization	14

5.2	Table 3: interest rates become increasingly tied to FICO and LTV	15
5.3	Tables 4 and 5: direct evidence on the disappearing role of hidden information . . .	15
5.4	Table 6: the interest rate becomes a worse predictor of default	16
5.5	Table 7 and Figure 1: the main failure of the historical default model	16
5.6	Table 8 and Figure 2: placebo inside the low-securitization period	16
5.7	Table 9 and Figures 3–6: where the model fails the most	18
5.8	Figure 7 and the investor discussion: house prices matter, but not enough	19
6	Short summary	20
7	Conclusion	21
7.1	Core conclusion (what the paper establishes)	21
7.2	Economic intuition	21
7.3	Takeaway	21

1 Big picture

- **Question.** Why did widely used statistical default models perform so poorly in the subprime mortgage boom? The paper asks whether the problem was not only bad macro shocks, but also a *Lucas critique*: securitization changed lender behavior and therefore changed the mapping from observables to defaults.
- **Setting.** The paper studies U.S. securitized subprime mortgages originated from 1997 to 2006. The central institutional change is the rise of “originate-and-distribute” lending, which increases the distance between the borrower and the ultimate investor.
- **Core conceptual idea.** Originators observe both *hard information* that can be reported to investors and *soft or otherwise unreported information* that is costly to verify or transmit. When securitization becomes easier, lenders have weaker incentives to collect and use the unreported information.
- **Main empirical challenge.** If default models break down over time, that could be because house prices collapse, because loan composition changes, or because the predictive relation itself changes. The paper’s job is to distinguish an incentive-induced regime change from ordinary forecasting error.
- **Main contribution.** The paper ties together three pieces of evidence:
 - interest rates increasingly depend only on reported hard information;
 - the interest rate becomes a worse predictor of actual default;
 - a default model estimated in the low-securitization era systematically underpredicts defaults in the high-securitization era, especially where soft information should matter most.
- **Main data design.** The core data are loan-level securitized non-agency mortgages from LoanPerformance, supplemented by industry-wide securitization rates and by an unusually rich single-lender data set from New Century Financial Corporation (NCFC) that contains internal ratings and rejected applications.
- **Headline market facts.** The number of securitized subprime loans in the sample rises from **24,067** in 1997 to **614,820** in 2006. The overall subprime securitization rate rises from about **37%** in 1997–2000 to **85%** in 2006.
- **Main pricing result.** In a yearly regression of mortgage interest rates on just FICO and LTV, the adjusted R^2 rises from about **9%** in 1997–2000 to about **47%** in 2006. Interest rates therefore become much more tightly pinned down by the variables reported to investors.
- **Main within-lender result.** At NCFC, the internal rating used by loan officers matters much less over time both for pricing and for acceptance decisions. That is direct evidence that unreported information is being downgraded in importance.
- **Main model-failure result.** A default model estimated on 1997–2000 loans underpredicts defaults every year from 2001 to 2006. Mean prediction error rises from about **4.0 percentage points** in 2001 to about **25.5 percentage points** in 2006.

- **Key cross-sectional result.** Prediction errors are larger for borrowers with **low FICO scores, high LTV ratios, low-documentation loans**, loans in **nonjudicial foreclosure states**, and low-documentation loans just **above the 620 FICO securitization threshold**.
- **Main credibility result.** The paper shows that the pattern is not just a generic misspecification problem:
 - a placebo test inside the low-securitization era produces little systematic error;
 - positive prediction errors already appear for 2001–2004 loans, before the national house-price collapse affects those vintages over the relevant 24-month horizon;
 - explicitly adding future house-price appreciation reduces but does not eliminate the breakdown of the model.
- **Economic takeaway.** Predictive risk models are endogenous to incentives. When securitization changes what lenders choose to observe and use, historical default models can become systematically wrong in exactly the regions of the credit distribution where soft information used to matter most.

2 Data and measurement (key objects)

2.1 Why this market is the right laboratory

- The subprime mortgage market relies heavily on statistical screening tools such as FICO scores, LTV ratios, and standardized loan characteristics.
- At the same time, originators can gather additional borrower information that is not easily transmitted to outside investors.
- The 1997–2006 period features a large and visible rise in securitization, so the paper can compare a low-securitization regime to a high-securitization regime.
- Mortgage performance can be tracked after origination, allowing the authors to compare ex ante pricing with ex post default outcomes.

Interpretation. This is exactly the setting where a Lucas critique should appear if the move from originate-and-hold to originate-and-distribute changes information production.

2.2 Primary data set: securitized subprime mortgages

The main data come from **LoanPerformance** and contain loan-level information on securitized non-agency mortgage loans.

Key facts about this data set:

- as of December 2006 it contains over **8,000** home-equity and nonprime loan pools;
- it includes about **16.5 million loans** in total and more than **\$1.6 trillion** in outstanding balances;

- vendor estimates suggest coverage of more than **90%** of securitized subprime loans by 2006.

The paper keeps **subprime** loans and excludes Alt-A. It focuses on conventional owner-occupied properties, mainly single-family homes, townhouses, and condominiums, and excludes unusual property categories and government-insured products.

Interpretation. The main sample is very strong for studying the *securitized* part of the market, but it does not directly observe all retained loans. That is why the paper supplements it with industry securitization rates and the NCFC lender file.

2.3 What the primary data observe

For each loan, the main data contain variables passed from the issuer to investors, including:

- loan amount and maturity;
- mortgage interest rate;
- borrower FICO credit score;
- loan-to-value ratio (LTV);
- mortgage type and loan purpose;
- documentation status;
- zip code and property characteristics.

Three measurement details matter:

- FICO is a hard-information summary of borrower credit history, mostly between 500 and 800 in this sample.
- LTV is both a collateral measure and, for borrowers without a second lien, a rough proxy for borrower wealth.
- Documentation is coded as full, limited, or no documentation; the paper combines limited and no documentation into **low-documentation loans**.

Important limitations:

- points and up-front fees are not observed;
- borrower variables are measured at origination only;
- cumulative LTV and second-lien information are unreliable in the early sample.

2.4 Secondary data: securitization rates and the NCFC loan file

The paper uses two additional data sources.

First, market-wide securitization rates. These come from *Inside B&C Lending*. They show a strong rise in the securitization rate of subprime loans:

- about **37%** in 1997–2000,
- **58%** in 2001,
- **76%** in 2004,
- **85%** in 2006.

Second, the New Century Financial Corporation data. These are unusually valuable because they include:

- accepted and rejected applications;
- loans later securitized and loans retained by the lender;
- internal variables that are *not* passed to investors.

Most important is an internal **rating** from 1 (best) to 20 (worst), assigned by NCFC loan officers. This rating is correlated with variables such as self-employment, marital status, referral status, and other debt, and is meant to summarize information visible to the lender but not necessarily to investors.

Interpretation. NCFC gives the paper a direct look at the unreported-information margin that is invisible in the market-wide securitized-loan sample.

2.5 Default and other core outcome variables

The paper defines

$$\text{Actual Default}_{it} = 1 \tag{1}$$

if loan i becomes at least **90 days delinquent within 24 months** of origination, and zero otherwise.

Why 24 months?

- FICO scores are designed to predict negative credit events over roughly a two-year horizon.
- A two-year horizon is typically before the first reset date of the most common ARMs in this market.

Other key outcomes are:

- interest rates at origination;
- acceptance or rejection of applications in NCFC data;
- prediction errors from historical default models.

2.6 Descriptive facts

- The number of securitized subprime loans rises more than fourfold between 2001 and 2006.
- The share of low-documentation loans rises from about **24.9%** in 1997 to **44.0%** in 2006.

- Mean LTV rises from about **80.5%** in 1997 to about **87.0%** in 2006.
- Mean FICO rises from about **611** in 1997 to about **636** in 2006.
- NCFC shows a similar securitization trend: from about **41%** in 1997–2000 to **96%** in 2006.

Interpretation. On the surface, the later-period loan pool does not look obviously weaker by the reported variables alone; if anything, FICO rises. That is exactly why a deterioration along the *unreported-information* margin is central to the paper.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: low- versus high-securitization regimes

The paper distinguishes between hard information X_{it} that is reported to investors and additional information Z_{it} that is observed by the lender but not transmitted to investors.

In the low-securitization regime, the lender both approves loans and prices them using both kinds of information:

$$A_{it} = f(X_{it}, Z_{it}), \quad (2)$$

$$r_{it} = g(X_{it}, Z_{it}). \quad (3)$$

In the high-securitization regime, costly unreported information becomes less valuable to collect and use, so the lender relies mainly on the information that investors see:

$$A_{it} = \tilde{f}(X_{it}), \quad (4)$$

$$r_{it} = \tilde{g}(X_{it}). \quad (5)$$

Interpretation. The core claim is not simply that loans get worse. It is that securitization changes the *decision rule* used by lenders, and therefore changes the relation between observables and loan quality.

3.2 Three empirical predictions of the framework

The framework implies three testable predictions.

- **Prediction 1:** Interest rates should depend more tightly on reported variables such as FICO and LTV as securitization rises.
- **Prediction 2:** The interest rate should become a worse predictor of default, because it no longer reflects as much soft information.
- **Prediction 3:** A default model estimated using the low-securitization mapping should underpredict defaults in the high-securitization period, especially for borrowers for whom soft

information matters more.

The paper’s simple two-type intuition is useful: when the lender stops collecting the extra signal Z , good and bad borrowers with the same hard information X are pooled more often, so the same interest rate now corresponds to a riskier average borrower.

3.3 Main interest-rate regression on hard information

The basic pricing regression is

$$r_i = \beta_0 + \beta_{\text{FICO}} \cdot \text{FICO}_i + \beta_{\text{LTV}} \cdot \text{LTV}_i + \varepsilon_i. \quad (6)$$

The paper estimates this equation year by year, using 1997–2000 as one combined low-securitization period and then separate yearly regressions from 2001 to 2006.

Interpretation. The coefficient values matter, but the key object is the **adjusted R^2** . A rising R^2 means that interest rates are increasingly determined by the variables that investors can see.

Main result:

- adjusted R^2 rises from **8.98%** in 1997–2000 to **46.67%** in 2006;
- the pattern remains when adding more contract variables, restricting to the largest lenders, restricting to fixed-rate loans, or using spreads instead of raw mortgage rates.

3.4 Shrinkage test for the distribution of interest rates

The paper also studies whether interest rates become less dispersed conditional on reported hard information. For a FICO-score bucket b in year t , it estimates

$$\sigma_{bt} = \alpha_b + \beta_b \cdot t + \varepsilon_{bt}, \quad (7)$$

where σ_{bt} is the standard deviation of interest rates within that FICO bucket.

Interpretation. If lenders stop using the hidden information margin, then borrowers with the same FICO score should receive increasingly similar interest rates.

Main result:

- for FICO 500–529, $\beta_b \approx -0.209$;
- for FICO 530–559, $\beta_b \approx -0.168$;
- shrinkage is much smaller for higher-FICO buckets.

This is exactly where soft information should matter most: among weaker hard-information borrowers.

3.5 NCFC pricing and acceptance tests using the internal rating

The NCFC data allow the paper to observe a lender-side internal rating that investors do not see. First, the paper re-estimates the interest-rate regression on NCFC first-lien loans. The adjusted R^2 of the base model rises from **10.8%** in 1997–2000 to **28.1%** in 2004.

Second, the paper adds the internal rating to the pricing regression. The extra explanatory power of the rating declines sharply:

- in 1997–2000, adding the rating lifts adjusted R^2 from **10.8%** to **16.3%**;
- by 2004, it lifts adjusted R^2 only from **28.1%** to **29.3%**;
- by 2006, only from **27.4%** to **28.0%**.

Third, the paper studies acceptance using

$$\Pr(\text{Accept}_{it} = 1) = \Phi(\beta_0 + \beta_{\text{Rating}} \cdot \text{Rating}_{it}). \quad (8)$$

The pseudo- R^2 of this logit falls from about **1.00%** in 1997–2000 to **0.09%** in 2006.

Interpretation. This is the paper’s cleanest micro evidence that a lender’s internal, unreported information becomes much less relevant as securitization intensifies.

3.6 Contemporaneous default regressions on the interest rate

To see whether pricing becomes less informative about ex post risk, the paper estimates year-by-year logits:

$$\Pr(\text{Actual Default}_{it} = 1) = \Phi(\beta_0 + \beta_r \cdot r_{it}). \quad (9)$$

It runs this separately for low-documentation and full-documentation loans.

Key results:

- for **low-documentation** loans, pseudo- R^2 falls from **3.42%** in 2001 to **1.12%** in 2004 and **0.65%** in 2006;
- for **full-documentation** loans, the same decline is visible later, falling to **0.92%** in 2005 and **0.38%** in 2006.

The paper also shows that at the lower part of the FICO distribution, a one-standard-deviation change in the interest rate implies a much smaller change in default risk in later years.

Interpretation. The interest rate remains statistically significant, but it becomes a much noisier signal of quality because it no longer reflects the unreported information margin.

3.7 Baseline default model estimated in the low-securitization regime

The central forecasting exercise estimates a baseline model on all securitized loans from 1997 to 2000:

$$\Pr(\text{Actual Default}_i = 1) = \Phi(\beta \cdot X_i + \beta^{\text{Low}} \cdot I_i^{\text{Low}} X_i). \quad (10)$$

Here:

- X_i includes the interest rate, FICO score, LTV ratio, an ARM dummy, and a prepayment-penalty dummy;
- I_i^{Low} indicates a low-documentation loan;
- the model also includes state fixed effects.

Interpretation. This is the paper’s stand-in for the kind of historical model a regulator or rating agency would use. The exact specification is less important than the fact that it is estimated on old data and does not account for changing incentives.

3.8 Out-of-sample prediction errors in the high-securitization period

Using fixed coefficients from the 1997–2000 model, the paper predicts default probabilities for loans issued in 2001–2006 and defines

$$\text{Prediction Error}_{it} = \text{Actual Default}_{it} - \widehat{\text{Predicted Default}}_{it}. \quad (11)$$

Then it studies both average prediction errors and their slope with respect to borrower observables using

$$\text{Prediction Error}_{it} = \alpha + \beta_{\text{FICO}} \cdot \text{FICO}_{it} + \beta_{\text{LTV}} \cdot \text{LTV}_{it}. \quad (12)$$

Main results:

- mean prediction error is positive every year: **3.96%** in 2001, **5.01%** in 2003, **7.79%** in 2004, **14.67%** in 2005, and **25.49%** in 2006;
- $\beta_{\text{FICO}} < 0$ and $\beta_{\text{LTV}} > 0$ every year, so underprediction is strongest for **low-FICO** and **high-LTV** borrowers.

Interpretation. The failure is not random. The model breaks in a way that matches the theory of lost soft information.

3.9 Rolling-window, placebo, and house-price-augmented designs

The paper adds three important design extensions.

Rolling window. It re-estimates the baseline using all information available up to each year. Errors become smaller, but remain positive and economically large.

Placebo in the low-securitization era. A model estimated on 1997–1998 and used to predict 1999–2000 low-documentation loans produces mean errors around **0.91%** and **0.97%**, not significantly different from zero, and no systematic FICO/LTV pattern.

House-price-augmented model. The paper adds realized two-year state house-price appreciation:

$$\text{HPA}_i = \frac{h_{s,t+2} - h_{s,t}}{h_{s,t}}. \quad (13)$$

This is a stringent robustness test because it gives the model future information that a regulator would not have had in real time.

Main result:

- for 2005 loans, mean error falls from about **8.3%** to **4.9%**;
- for 2006 loans, from about **15.1%** to **6.1%**;
- for 2001–2004 loans, positive errors remain and often change very little.

Interpretation. House prices matter, especially late in the boom, but they do not explain the whole failure of the model.

4 Identification and empirical design (what makes the estimates credible)

4.1 What the paper identifies

The paper is not trying to estimate a single clean treatment effect of securitization on default. Instead, it tests whether a rise in securitization changes the *data-generating process* linking observables to defaults.

That makes the design more diagnostic than purely causal:

- do pricing equations increasingly load on reported hard information?
- do interest rates lose forecasting power?
- do historical default models break in systematic directions?

Interpretation. The core object is the *stability* of the predictive relation, not only the level of default.

4.2 Why the pricing evidence is not just lender composition

A natural concern is that later years simply contain different lenders, not different behavior by the same lenders.

The paper answers this in several ways:

- the rise in adjusted R^2 remains when restricting to the largest lenders that make roughly **80%** of the loans;
- the same pattern appears for fixed-rate mortgages only;
- the same pattern appears when using mortgage spreads rather than raw rates;
- the single-lender NCFC file shows the same change *within one lender*.

Interpretation. The results look like a change in lender behavior, not just a changing mix of lenders.

4.3 Why the story is about lost information, not FICO becoming more informative

Another alternative is that the market simply learned to use FICO better over time.

The paper shows the opposite. When it regresses default within 24 months on the FICO score alone, the pseudo- R^2 *falls*, from about **5%** in 1997–2000 to about **0.01%** in 2006.

Interpretation. Hard information is becoming a worse standalone predictor of default, even while lenders rely on it more. That is exactly what one would expect if the pool deteriorates along an unreported dimension.

4.4 Why the forecasting failure is not just generic misspecification

The placebo test is crucial here. When the baseline model is estimated inside one low-securitization regime and used to predict nearby years in that same regime, the average error is near zero and has no strong systematic FICO or LTV gradient.

Interpretation. The default model is not “bad” in a timeless sense. It fails when the environment changes.

4.5 Cross-sectional designs that isolate the information channel

The paper adds three strong heterogeneity designs.

Low-documentation versus full-documentation loans. Because more information remains unreported on low-documentation loans, prediction errors should be larger there. Using rolling windows, that is exactly what happens, especially late in the sample:

- 2005 mean error: **10.58%** for low-documentation versus **5.85%** for full-documentation;
- 2006 mean error: **20.11%** versus **9.84%**.

Judicial versus nonjudicial foreclosure states. Using matched counties across state borders, the paper argues securitization is easier in nonjudicial states. Prediction errors are then larger in those states; the overall mean error is about **0.122** versus **0.100** in judicial states.

The 620 FICO securitization threshold. Among low-documentation loans, those just above 620 are more easily securitized than those just below it. Mean prediction error is about **18.2%** above 620 versus **16.6%** below 620.

Interpretation. These tests matter because they compare loans that are much closer on observables and local macro conditions, but differ in the ease of securitization.

4.6 Alternative explanations the paper addresses

The paper directly discusses several alternative mechanisms.

Falling house prices. This cannot explain the whole pattern because:

- positive prediction errors already show up for 2001–2004 loans;
- cross-sectional tests compare loans facing similar local conditions;
- explicit house-price controls reduce but do not eliminate the errors.

Lower cost of capital. A pure lower-cost-of-capital story could bring in riskier borrowers, but it would tend to *widen* the dispersion of interest rates within a FICO bucket. The paper finds the opposite: dispersion *shrinks*, especially at low FICO.

Second-lien borrowing and CLTV issues. The data are imperfect on this, but available robustness checks do not overturn the underprediction results.

Standardization of contract terms. Standardization alone cannot explain why shrinkage in rate dispersion is strongest exactly where soft information should matter most, nor why the mapping from observables to defaults changes in the documented direction.

4.7 Role of investors and limits of interpretation

The paper is largely agnostic on whether investors themselves were rationally fooled. The theoretical mechanism works even with rational investors because the key friction is that soft information cannot be contracted upon.

The paper does, however, offer suggestive evidence that rating models and tranche protection did not fully adjust:

- for 2005–2006 pools with many low-documentation loans, the relation between AAA subordination levels and model-predicted risk is weak at best;
- this lines up with the paper’s broader message that investors and rating agencies did not fully internalize the regime change.

Main limits:

- the primary sample is only the securitized part of the subprime market;
- investor-pricing data are indirect;

- the paper is strongest as a diagnostic of model instability, not as a full structural decomposition of all crisis channels.

4.8 Relation to the literature

The paper sits at the intersection of three literatures:

- the **Lucas critique**, because the mapping from observables to outcomes changes when incentives change;
- **securitization and screening**, alongside work showing that easier securitization weakens underwriting;
- **credit-risk modeling and regulation**, because ratings, capital rules, and loan screens often rely on historical statistical models.

Interpretation. The paper’s distinctive contribution is not merely “securitization worsened loans.” It is that securitization changed the *meaning* of observed borrower characteristics for anyone using backward-looking statistical models.

5 Tables and figures

5.1 Tables 1 and 2: market growth and securitization

Table 1

Summary statistics, primary data set.

This table reports summary statistics of FICO scores, loan-to-value (LTV) ratios and information on the documentation reported by the borrower (full, limited, or no) when taking the loan. Full-documentation loans provide verification of income as well as assets of the borrower. Limited documentation provides no information about the income but does provide some information about the assets. No documentation loans provide no information about income or assets. We combine limited and no documentation loans and call them “low-documentation loans.”

Origination year	Number of loans	Proportion with low documentation (percent)	Mean LTV ratio (percent)	Mean FICO score
1997	24,067	24.9	80.5	611
1998	60,094	23.0	81.5	605
1999	104,847	19.2	82.2	610
2000	116,778	23.5	82.3	603
2001	136,483	26.0	84.6	611
2002	162,501	32.8	85.6	624
2003	318,866	38.9	87.0	637
2004	610,753	40.8	86.6	639
2005	793,725	43.4	86.3	639
2006	614,820	44.0	87.0	636

Table 2

Securitization rate over time.

This table reports the securitization rate for the overall subprime mortgage market and for New Century Financial Corporation (NCFC). The yearly securitization proportion for the overall market is obtained from *Inside B&C Lending*, a publication that has extensive coverage of the subprime mortgage market. Data on NCFC securitization rates comes from the origination and servicing loan files that encompass all lending activities of NCFC from 1997 to 2007.

Origination year	Securitization rate (percent)						
	1997–2000	2001	2002	2003	2004	2005	2006
Overall market	37	58	62	66	76	79	85
NCFC loans	41	50	77	88	92	85	96

Read:

- Table 1 shows huge growth in securitized subprime issuance and a steady rise in low-documentation lending.
- Reported borrower quality does not collapse: mean FICO actually rises from 611 to 636.

Table 3
 Reliance of interest rates on FICO scores and loan-to-value (LTV) ratios.
 This table reports estimates from the yearly regression of interest rates on FICO and LTV, using our primary data set. Standard errors are in parentheses.
 *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

Origination year	Base model coefficients		Number of observations	Adjusted R^2 (percent) of various models		
	β_{FICO}	β_{LTV}		Base model	With additional contract variables	Including only lenders making 80% of loans
1997–2000	−0.009*** (0.0001)	0.033*** (0.0003)	305,786	8.98	11.38	8.40
2001	−0.012*** (0.0001)	0.038*** (0.0004)	136,483	19.49	22.74	20.13
2002	−0.011*** (0.0001)	0.071*** (0.0001)	162,501	17.42	26.43	15.66
2003	−0.012*** (0.0001)	0.079*** (0.0001)	318,866	29.72	41.26	33.29
2004	−0.010*** (0.0001)	0.097*** (0.0001)	610,753	36.85	45.39	41.00
2005	−0.009*** (0.0001)	0.110*** (0.0001)	793,725	43.91	50.14	52.82
2006	−0.011*** (0.0001)	0.115*** (0.0001)	614,820	46.67	50.83	46.72

- Table 2 shows the securitization regime change directly: overall securitization rises from about 37% in 1997–2000 to 85% in 2006, and NCFC shows a similar pattern.

Economic message. The boom is not just more of the same market. The environment in which loans are originated changes sharply.

5.2 Table 3: interest rates become increasingly tied to FICO and LTV

Read:

- The adjusted R^2 of the base interest-rate regression jumps from 8.98% to 46.67%.
- This rise survives extra controls and lender restrictions.
- The coefficients on FICO stay negative and on LTV stay positive, but the main result is the rise in explanatory power.

Economic message. Pricing increasingly reflects the variables investors observe, which is exactly what the theory predicts when lenders stop relying on hidden information.

5.3 Tables 4 and 5: direct evidence on the disappearing role of hidden information

Read:

- Table 4 shows that within-FICO interest-rate dispersion shrinks over time, most strongly at low FICO scores.
- Table 5 shows that inside NCFC the internal lender rating matters less and less for both pricing and acceptance.
- In Panel B of Table 5, the pseudo- R^2 of acceptance on the internal rating falls from about 1.00% to 0.09%.

Economic message. These are the paper’s strongest pieces of evidence that the unreported-information margin is being abandoned.

Table 4

Shrinkage in the distribution of interest rates.

We report estimates from a regression of yearly standard deviation of interest rates at different FICO scores on time. The regressions are estimated separately in buckets of 30 FICO points, starting with the bucket 500–529 and ending with 770–799. We include loans originated between 1997 and 2006. Standard errors are in parentheses. *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

FICO score bucket	β_b	Standard error	R^2 (percent)
500–529	–0.209***	0.044	70.7
530–559	–0.168***	0.027	81.1
560–589	–0.099***	0.026	59.3
590–619	–0.042	0.025	17.0
620–649	–0.028	0.019	12.4
650–679	–0.055**	0.021	38.7
680–709	–0.058***	0.020	45.9
710–739	–0.079***	0.025	49.0
740–769	–0.085***	0.029	45.9
770–799	–0.065***	0.013	73.2

Table 5

Reliance of interest rates on reported and unreported variables.

This table reports results from the New Century Financial Corporation sample. For NCTC, we have information on accepted and rejected loan applications and on variables reported to investors as well as variables that are collected by the lender but not reported to investors. Panel A shows the coefficients and adjusted R^2 from an OLS regression of interest rates on the FICO score and LTV ratio and (last column) the internal rating measure. Panel B shows the coefficients from a logistic regression of the accept or reject decision for a loan application on the internal rating measure. Standard errors are in parentheses. *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

Panel A: OLS regression of interest rate on FICO score and LTV ratio					
Origination year	Base model coefficients		Number of observations	Adjusted R^2 (percent)	
	β_{FICO}	β_{LTV}		Base model	Model including internal rating
1997–2000	–0.0053*** (0.0001)	0.014*** (0.0008)	21,553	10.8	16.3
2001	–0.0072*** (0.0002)	0.013*** (0.0016)	7,302	12.9	18.9
2002	–0.0084*** (0.0001)	0.009*** (0.0010)	15,092	19.5	24.5
2003	–0.0085*** (0.0001)	0.020*** (0.0006)	33,690	25.1	28.6
2004	–0.0075*** (0.0001)	0.019*** (0.0005)	63,174	28.1	29.3
2005	–0.0062*** (0.0001)	0.060*** (0.0005)	84,002	23.9	24.4
2006	–0.0064*** (0.0001)	0.066*** (0.0005)	82,163	27.4	28.0
Panel B: Logit regression of accept/reject decision on internal rating measure					
Origination year	β_{Rating}	Number of observations	Pseudo- R^2 (percent)		
1997–2000	–0.053*** (0.002)	60,049	1.00		
2001	–0.059*** (0.004)	14,905	1.12		
2002	–0.070*** (0.003)	29,656	1.08		
2003	–0.097*** (0.004)	71,188	0.76		
2004	–0.075*** (0.004)	154,893	0.21		
2005	–0.080*** (0.004)	199,369	0.16		
2006	–0.056*** (0.004)	210,856	0.09		

5.4 Table 6: the interest rate becomes a worse predictor of default

Read:

- The pseudo- R^2 in the default-on-interest-rate logit falls strongly over time, especially for low-documentation loans.
- The marginal default sensitivity to a one-standard-deviation change in rates also declines.

Economic message. The interest rate is still informative, but much less so, because it no longer reflects as much private information about borrower quality.

5.5 Table 7 and Figure 1: the main failure of the historical default model

Read:

- Table 7 shows positive mean prediction errors in every year from 2001 to 2006.
- The prediction error is more positive for low-FICO and high-LTV borrowers.
- Figure 1 shows the entire distribution of mean prediction errors shifting to the right over time.

Economic message. The failure is systematic, not accidental. Historical models estimated in the low-securitization era increasingly underpredict true default risk as securitization rises.

5.6 Table 8 and Figure 2: placebo inside the low-securitization period

Read:

- The model estimated on 1997–1998 predicts 1999–2000 defaults reasonably well.

Table 6

Contemporaneous default regressions.

This table reports the coefficients and pseudo- R^2 from a logistic regression of actual defaults on loan interest rates. A loan is defined to be in default if it is delinquent for at least 90 days within 24 months from the year of origination. Standard errors are in parentheses. *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

Panel A: Low-documentation loans				
Origination year	β_r	Constant (β_0)	Pseudo- R^2 (percent)	Number of observations
1997–2000	0.282*** (0.00920)	−4.996*** (0.0965)	2.43	65,895
2001	0.333*** (0.0112)	−5.159*** (0.113)	3.42	35,110
2002	0.224*** (0.00709)	−4.079*** (0.0689)	2.54	52,967
2003	0.224*** (0.00514)	−4.023*** (0.0442)	2.21	123,766
2004	0.159*** (0.00341)	−3.215*** (0.0282)	1.12	248,839
2005	0.127*** (0.00247)	−2.331*** (0.0208)	0.73	343,581
2006	0.111*** (0.00231)	−1.444*** (0.0215)	0.65	270,284

Panel B: Full-documentation loans				
Origination year	β_r	Constant (β_0)	Pseudo- R^2 (percent)	Number of observations
1997–2000	0.211*** (0.00376)	−4.065*** (0.0409)	1.94	231,103
2001	0.243*** (0.00506)	−4.051*** (0.0534)	2.61	98,751
2002	0.177*** (0.00437)	−3.344*** (0.0422)	1.88	107,648
2003	0.240*** (0.00355)	−3.856*** (0.0307)	2.93	194,010
2004	0.199*** (0.00261)	−3.268*** (0.0212)	1.83	360,646
2005	0.140*** (0.00215)	−2.451*** (0.0177)	0.92	448,422
2006	0.0858*** (0.00216)	−1.689*** (0.0199)	0.38	343,393

Table 7

Default model—failing to predict failure.

We report estimates from a baseline default model estimated for low and full-documentation loans originated from 1997 to 2000 in Panel A. A loan is defined to be in default if it is delinquent for at least 90 days within 24 months from the year of origination. Panel B reports the β coefficients from a regression of prediction error on FICO score and LTV ratio for loans issued from each year 2001 to 2006, and also reports the mean prediction errors for each vintage. *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

Panel A: Coefficients of baseline model in low securitization regime, 1997–2000							
FICO	r	LTV	β^{FICO}_x	β^{LTV}_x	β^{FICO}_x	Pseudo R^2 (percent)	Number of observations
−0.009*** (0.0001)	0.231*** (0.006)	0.002*** (0.001)	0.001*** (0.0001)	−0.043*** (0.016)	−0.006*** (0.001)	7.05	267,511

Panel B: Prediction errors during high securitization regime.							
Origination year	Actual and predicted defaults		Regression of prediction error on FICO, LTV				
	Mean prediction error (percent)	Actual defaults (percent)	β_{FICO} ($\times 10^{-3}$)	β_{LTV} ($\times 10^{-2}$)	Number of observations	Adjusted R^2 (percent)	
2001	3.96***	16.0	−0.123***	0.052*** (0.018)	128,772 (0.010)	0.05	
2002	4.70***	14.1	−0.197***	0.082*** (0.015)	152,057 (0.010)	0.15	
2003	5.01***	11.9	−0.428***	0.077*** (0.010)	308,340 (0.010)	0.61	
2004	7.70***	13.9	−0.621***	0.081*** (0.008)	596,485 (0.004)	0.97	
2005	14.67***	21.1	−1.341***	0.143*** (0.030)	788,299 (0.007)	3.90	
2006	25.49***	33.2	−1.120***	0.190*** (0.012)	608,559 (0.005)	1.60	

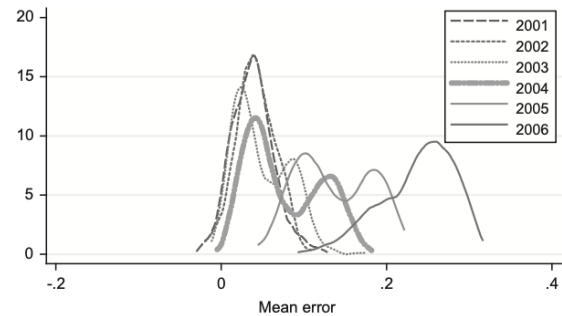


Fig. 1. Kernel density of mean prediction errors over time, all loans. Kernel density of mean prediction errors (*Actual Defaults* − *Predicted Defaults*) of a baseline model estimated for loans issued in 1997 to 2000. For each subsequent year, we first determine the mean prediction error at each FICO score and then plot the kernel density of the mean errors. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991).

Table 8

Default model—placebo test.

We report estimates from a baseline default model estimated for low-documentation loans issued in 1997 and 1998 in Panel A. A loan is defined to be in default if it is delinquent for at least 90 days within 24 months from the year of origination. Panel B reports the β coefficients from a regression of prediction error on FICO score and LTV ratio for loans issued in 1999 and 2000, and also reports the mean prediction errors for each vintage. *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

Panel A: Coefficients of baseline model in low securitization regime					
Origination years	FICO	r	LTV	Pseudo- R^2 (percent)	Number of observations
1997–1998	–0.009*** (0.0005)	0.249*** (0.034)	–0.008*** (0.003)	8.11	16,002
1997–1999	–0.007*** (0.003)	0.259*** (0.022)	–0.003* (0.001)	7.94	33,868

Panel B: Prediction errors during high securitization regime.					
Origination year	Actual and predicted defaults		Regression of prediction error on FICO, LTV		
	Mean prediction error (percent)	Actual defaults (percent)	β_{FICO} ($\times 10^{-2}$)	β_{LTV} ($\times 10^{-2}$)	Adjusted R^2 (percent)
1999	0.91	11.0	0.039 (0.038)	0.026 (0.023)	0.01
2000	0.97	11.9	0.039 (0.034)	–0.026 (0.020)	0.01

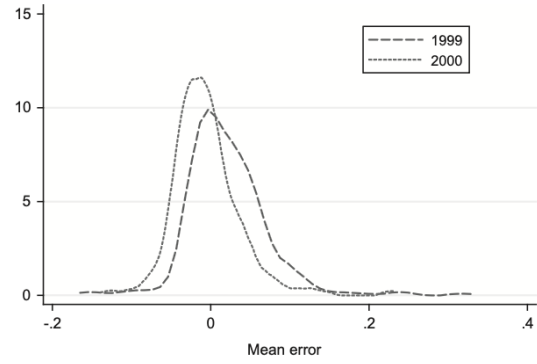


Fig. 2. Placebo test—mean prediction errors in low securitization regime. This figure presents the Epanechnikov kernel density of mean prediction errors (*Actual Defaults* – *Predicted Defaults*) for low-documentation loans issued in 1999 to 2000. The baseline model for 1999 loans is estimated over 1997 and 1998 and the baseline model for 2000 loans is estimated from 1997 through 1999. For each year 1999 and 2000, we first determine the mean prediction error at each FICO score and then plot the kernel density of the mean errors. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991).

Table 9

Mean prediction errors for low- and full-documentation loans.

We report the mean prediction errors for low- and full-documentation loans issued from 2001 through 2006. The estimation uses a rolling window approach with separate baseline models for low-documentation and full-documentation loans. That is, the predictions for year $t+1$ are based on a model estimated over the years 1 through t , where year 1 is 1997. *** indicates significance at the 1% level; ** at the 5% level; * at the 10% level.

Origination year	Low-documentation (percent)	Full-documentation (percent)	Difference (percent)
2001	3.40	3.80	–0.40
2002	2.78	2.79	–0.01
2003	3.20	2.21	0.99***
2004	5.17	3.51	1.66***
2005	10.58	5.85	4.73***
2006	20.11	9.84	10.27***

- Average errors are close to zero and do not show the same FICO/LTV gradients.
- Figure 2 shows the error distribution centered around zero rather than shifted right.

Economic message. The breakdown is tied to the regime change, not to a fundamentally unusable modeling strategy.

5.7 Table 9 and Figures 3–6: where the model fails the most

Read:

- Figure 3 and Table 9 show larger errors for low-documentation loans than for full-documentation loans.
- Figures 4 and 5 show stronger model failure in nonjudicial foreclosure states, where securitization appears easier.

- Figure 6 shows larger errors for low-documentation loans just above the 620 securitization threshold than just below.

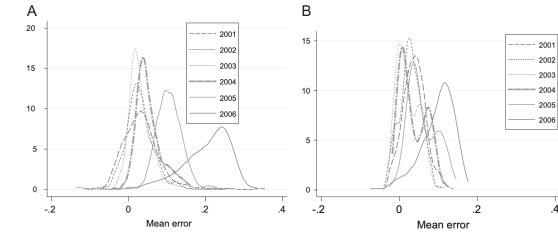


Fig. 3. Mean prediction errors for low- and full-documentation loans. This figure presents the Epanechnikov kernel density of mean prediction errors ($\text{Actual Defaults} - \text{Predicted Defaults}$) on low-documentation (Panel A) and full-documentation (Panel B) loans of a baseline model using a rolling estimation window. The prediction errors for year $t+1$ are from a baseline model estimated over 1997 to year t . For each year, we first determine the mean prediction error at each FICO score and then plot the kernel density of the mean errors. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991).

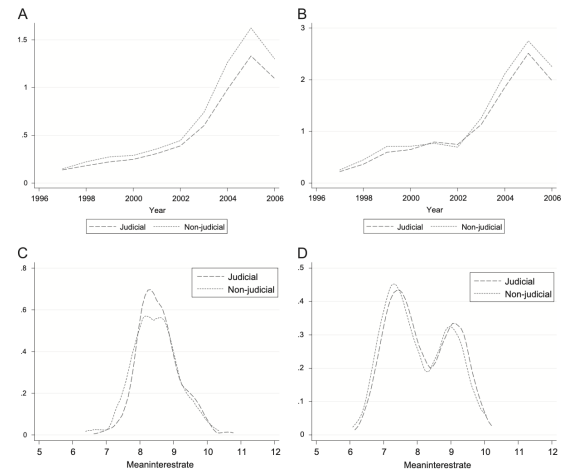


Fig. 4. Loans in judicial and nonjudicial foreclosure states. The average annual number of newly-issued securitized loans per thousand households are shown for low-documentation loans (Panel A) and full-documentation loans (Panel B) in states that require judicial foreclosures and those that allow nonjudicial foreclosures. The Epanechnikov kernel densities of interest rates are shown for low-documentation loans (Panel C) and full-documentation loans (Panel D) in states that require judicial foreclosures and those that allow nonjudicial foreclosures. To generate the kernel densities, we first average the interest rate at each FICO score. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991).

Economic message. Every heterogeneity test points in the same direction: failure is most severe exactly where hidden information should be most valuable or securitization should be easiest.

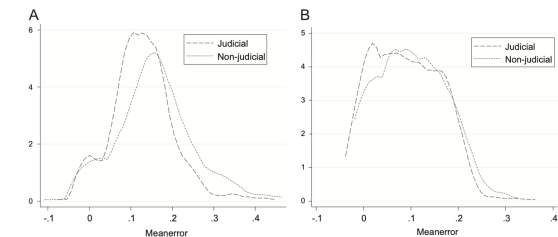


Fig. 5. Mean prediction errors for loans in judicial and nonjudicial foreclosure states. This figure shows the Epanechnikov kernel density of mean prediction errors ($\text{Actual Defaults} - \text{Predicted Defaults}$) of a baseline model estimated for loans issued in 1997 to 2000. The model is estimated separately for low- and full-documentation loans in states with judicial and nonjudicial foreclosures. For each subsequent year, we first determine the mean prediction error at each FICO score. We then average the errors across years at each FICO score and plot the kernel density of the mean errors. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991). (Panel A) Low-documentation loans and (Panel B) Full-documentation loans.

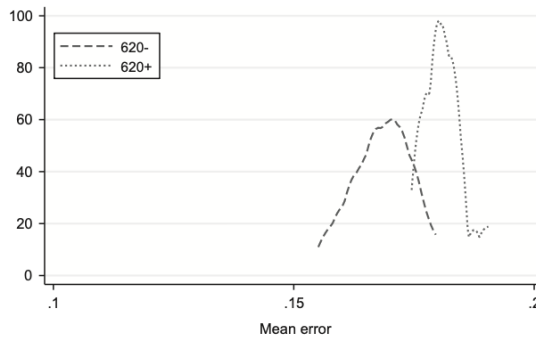


Fig. 6. Mean prediction errors for 620+ and 620- low-documentation loans. This figure presents the Epanechnikov kernel density of mean prediction errors ($\text{Actual Defaults} - \text{Predicted Defaults}$) for low-documentation loans issued in 1999 to 2000. The dashed line represents the prediction errors on loans with FICO scores from 621 to 630 (i.e., 620+ loans) and the dotted line the prediction errors on loans with FICO scores from 610 to 619 (i.e., 620- loans). We determine the prediction errors across loans at each FICO score and then plot the kernel density of the mean errors. The prediction errors are relative to a baseline model estimated on loans issued in the years 1997 through 2000. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991).

5.8 Figure 7 and the investor discussion: house prices matter, but not enough

Read:

- Adding realized future house-price appreciation shifts the prediction-error distributions left for

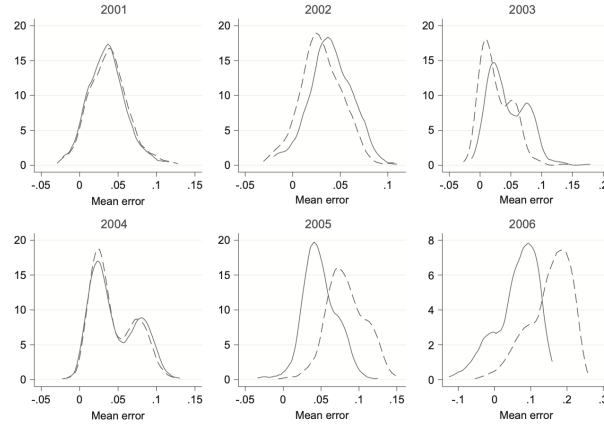


Fig. 7. Explicitly incorporating house price effects. This figure presents the Epanechnikov kernel density of mean prediction errors ($\text{Actual Defaults} - \text{Predicted Defaults}$) on all loans of a baseline model using a rolling estimation window. The prediction errors for year $t+1$ are from a baseline model estimated over 1997 to year t , with and without including house price appreciation (HPA) as an explanatory variable. For each year, we first determine the mean prediction error at each FICO score and then plot the kernel density of the mean errors. For each year, the dashed line represents the density of errors without HPA; the solid line, the density of errors with HPA included. The bandwidth for the density estimation is selected using the plug-in formula of Sheather and Jones (1991).

2005 and 2006, but they stay mostly positive.

- Earlier years change little.
- The later discussion on AAA subordination levels suggests that even investors and rating technology did not fully adjust to the regime change.

Economic message. The crisis cannot be reduced to house prices alone. The predictive relation itself had changed before the crash fully arrived.

6 Short summary

- Securitization increases the distance between originators and investors and weakens the incentive to collect soft information.
- As a result, interest rates increasingly depend only on reported hard information such as FICO and LTV.
- The interest rate therefore becomes a worse predictor of ex post default.
- A historical default model estimated in 1997–2000 systematically underpredicts defaults in 2001–2006.
- The underprediction is strongest for low-FICO, high-LTV, low-documentation, and more-easily-securitized loans.
- Placebo tests and house-price robustness checks show that the result is a regime-change problem, not just a generic forecasting failure.
- The broader lesson is that regulation and rating systems based on backward-looking statistical models can be undermined when market structure changes agents' incentives.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the poor performance of subprime default models was not only a story about bad macro realizations. A major part of the failure came from a shift in lender incentives caused by securitization, which changed the relation between observed borrower characteristics and true loan quality.

7.2 Economic intuition

When investors pay only for what they can verify, lenders stop investing in what investors cannot verify. Hard information gets more weight, soft information gets less weight, and a model built on older data becomes misleading because the approval and pricing process has changed.

7.3 Takeaway

This paper shows that predictive models are not policy-invariant. Once incentives change, the meaning of observables changes too. That lesson applies far beyond subprime mortgages: any regulatory, rating, or risk-management model that ignores endogenous behavioral responses is vulnerable to the same kind of failure.